

The big idea. Every function is built from simpler pieces. Each rule tells you how to differentiate one way of *combining* pieces: multiplying (product), dividing (quotient), or plugging one into another (chain). The real skill is seeing the structure.

WHICH RULE FIRST?

The **last** thing that happens when you plug in a number is the **first** rule you use.

FUNCTION	PLUG IN A NUMBER: WHAT DO YOU DO LAST?	RULE TO USE FIRST
$x^3 + \sin x$	Add	Sum / difference
$5 \cos x$	Multiply by the constant 5	Constant multiple
$x^2 \sin x$	Multiply x^2 by $\sin x$	Product
$\frac{x^2 + 1}{x - 3}$	Divide the top by the bottom	Quotient
$(3x^2 - 5)^4$	Raise to the 4th power	Chain
$\sin 4x$	Take the sine	Chain

THE FIVE STEPS

- 1 Find the last thing that happens. That is your first rule.
- 2 Name the pieces. f and g , or outside and inside.
- 3 Differentiate each piece on the side. If a piece has its own structure, repeat Steps 1-3 on it.
- 4 Assemble using the rule's pattern.
- 5 Simplify. Factor out common factors.

Top three mistakes: (1) $(fg)' \neq f'g'$. (2) Quotient rule order: the *top's* derivative comes first, $f'g - fg'$. (3) Forgetting the inside: $(\sin 4x)' = 4 \cos 4x$, not $\cos 4x$.

FORMULA BANK

BASIC RULES	TRIG	EXPONENTIAL & LOG
Constant $(c)' = 0$	$(\sin x)' = \cos x$ $(\cos x)' = -\sin x$	$(e^x)' = e^x$ $(a^x)' = a^x \ln a$
Constant multiple $(cf)' = c f'$	$(\tan x)' = \sec^2 x$ $(\cot x)' = -\csc^2 x$	$(\ln x)' = \frac{1}{x}$ $(\log_a x)' = \frac{1}{x \ln a}$
Sum / difference $(f \pm g)' = f' \pm g'$	$(\sec x)' = \sec x \tan x$ $(\csc x)' = -\csc x \cot x$	
Power $(x^n)' = nx^{n-1}$		

THE BIG THREE	HANDY POWERS
Product $(fg)' = f'g + fg'$ first' · second + first · second'	Square root $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
Quotient $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ “lo d-hi minus hi d-lo, over lo lo”	Reciprocal $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$
Chain $[f(g(x))]' = f'(g(x)) \cdot g'(x)$ outside' (inside kept) × inside'	Rewrite first $\sqrt[n]{x^m} = x^{m/n}$
Chain, Leibniz $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ where u is the inside	

ONE RULE AT A TIME

Product $y = x^2 \sin x$ PIECES $f = x^2, g = \sin x$ DIFF. $f' = 2x, g' = \cos x$ ASSEMBLE $y' = 2x \sin x + x^2 \cos x$ SIMPLIFY $y' = x(2 \sin x + x \cos x)$	Quotient $y = \frac{x^2 + 1}{x - 3}$ PIECES $f = x^2 + 1, g = x - 3$ DIFF. $f' = 2x, g' = 1$ ASSEMBLE $y' = \frac{2x(x - 3) - (x^2 + 1)(1)}{(x - 3)^2}$ SIMPLIFY $y' = \frac{x^2 - 6x - 1}{(x - 3)^2}$	Chain $y = (3x^2 - 5)^4$ PIECES outside u^4 , inside $u = 3x^2 - 5$ DIFF. outside' $4u^3$, inside' $6x$ ASSEMBLE $y' = 4(3x^2 - 5)^3 \cdot 6x$ SIMPLIFY $y' = 24x(3x^2 - 5)^3$
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COMBINATIONS: RULES INSIDE RULES

Each tree shows the structure. Read it top-down: the top box is the rule you apply first, and any piece with its own box needs its own rule in Step 3.

Walkthrough 1: Product with a chain inside $y = x^3 \sqrt{2x+1}$

- 1 LAST OP** Multiply x^3 by $\sqrt{2x+1}$, so the **product rule** comes first.
- 2 PIECES** $f = x^3, g = \sqrt{2x+1} = (2x+1)^{1/2}$ (rewrite roots as powers)
- 3 DIFF.** $f' = 3x^2$. g is a chain: $g' = \frac{1}{2}(2x+1)^{-1/2} \cdot 2 = (2x+1)^{-1/2}$
- 4 ASSEMBLE** $y' = f'g + fg' = 3x^2(2x+1)^{1/2} + x^3(2x+1)^{-1/2}$
- 5 SIMPLIFY** Factor out the lowest power of each factor, $x^2(2x+1)^{-1/2}$:

$$y' = x^2(2x+1)^{-1/2} [3(2x+1) + x] = \frac{x^2(7x+3)}{\sqrt{2x+1}}$$

Walkthrough 2: Quotient with a chain inside $y = e^{2x} / (x^2 + 1)$

- 1 LAST OP** Divide the top by the bottom, so the **quotient rule** comes first.
- 2 PIECES** top $f = e^{2x}$, bottom $g = x^2 + 1$
- 3 DIFF.** f is a chain: $f' = e^{2x} \cdot 2 = 2e^{2x}$. $g' = 2x$
- 4 ASSEMBLE** $y' = \frac{f'g - fg'}{g^2} = \frac{2e^{2x}(x^2 + 1) - e^{2x}(2x)}{(x^2 + 1)^2}$
- 5 SIMPLIFY** Factor $2e^{2x}$ from the top: $y' = \frac{2e^{2x}(x^2 - x + 1)}{(x^2 + 1)^2}$

Walkthrough 3: A chain inside a chain $y = \sin^2(4x)$

- 1 LAST OP** $\sin^2(4x)$ means $(\sin 4x)^2$. Squaring happens last, so the **chain rule** with outside $()^2$.
- 2 PIECES** outside u^2 , inside $u = \sin 4x$, which is *another chain*
- 3 DIFF.** outside' $= 2u$. Inside': $(\sin 4x)' = \cos 4x \cdot 4$
- 4 ASSEMBLE** $y' = 2 \sin 4x \cdot 4 \cos 4x$ (one factor per layer of the tree)
- 5 SIMPLIFY** $y' = 8 \sin 4x \cos 4x$ (or $4 \sin 8x$ with the double-angle identity)

Peeling the onion: in a chain of chains, write one derivative per layer, working from the outside in, and multiply them all: $[f(g(h(x)))]' = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$.